

IIIT-PGEE



General Aptitude QUANTITATIVE APTITUDE

Lecture No.- 03



By- Amulya Ratan Sir

Recap of Previous Lecture



Topics

✓
Averages & Percentages



Topics to be Covered



Topic-1

Percentages ✓

Topic-2

Profit & Loss ✓

Topic-3

Mixtures Alligations

Questionnaire:



$$1:01\frac{9}{11}$$

$$1:09\frac{1}{11}$$

Assignment

#Q. At what time between 1 and 2 o'clock do the hands of a clock subtend an angle of 20° ?

$$\frac{10}{5.5} = \frac{20}{11} = 1\frac{9}{11} \text{ min}$$

$$\frac{50}{5.5} = \frac{100}{11} = 9\frac{1}{11} \text{ min}$$



Thursday

22nd March, 2040

$$6+5=11 \rightarrow \text{odd}$$

$$\begin{array}{r} 4 \\ \times 9 \\ \hline 36 \end{array}$$

$$\boxed{\begin{array}{r} 2000 \rightarrow 0 \\ 39 \rightarrow 6 \end{array}}$$

$$L: 4 \quad N: 4$$

$$9 \quad 30$$

$$\textcircled{18} \quad \textcircled{30}$$

$$4 + 2 = 6$$

$$\frac{2039 \rightarrow \textcircled{6}^e}{}$$

$$J - 3$$

$$F - 1$$

$$M - 1 \quad (22)$$

$$\frac{\quad}{\textcircled{5}}$$

[MCQ]



#Q. The average of 5 consecutive integers starting with x is y . What is the average of 6 consecutive numbers starting with $(x+2)$?

Assignment

A $y + 3$

B $\frac{2y + 9}{2}$

C $y + 2$

D $\frac{2y + 5}{2}$

$$x, x+1, x+2, x+3, x+4 \Rightarrow y$$

$$y, y+1, y+2, y+3, y+4, y+5 \Rightarrow$$

$$\frac{y+2 + y+3}{2} = \frac{2y+5}{2}$$

[MCQ]



$$A = \frac{S}{No} \Rightarrow A \times No. = S \quad A = 130$$

#Q. A cricketer has certain average of runs for his 64 innings. In his 65th innings, he is bowled out for no score on his part. This brings down his average by 2 runs. His new average is?

Assignment

$$\rightarrow 130 - 2 = 128$$

A 130

B 128

C 70

D 68

$$A - 2 = \frac{64A + 0}{65}$$

$$\Rightarrow 65A - 130 = 64A$$

$$A = 130$$

[MCQ]

$$\% \text{ error} = \left| \frac{\text{diff}}{\text{A.V}} \right| \times 100$$



#Q. The population of a town is increasing at a uniform rate. If its population was 90,000 and 96,000 in 2022 and 2023 respectively, what would be its population in 2024?

Assignment

A 1,02,000

B 1,02,400

C 1,02,720

D 1,02,960

$$\begin{array}{r} 32 \\ 96000 \\ \hline 90,000 \end{array} \times \begin{array}{r} 3200 \\ 96000 \\ \hline \end{array}$$

$$= 1,02,400$$

[MCQ]



#Q. 8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25 actually voted. In that election number of person between 20 and 25, who actually voted, was what percentage of those eligible to vote?

Assignment

A 4.2%

B 8%

C 6.4%

D 6.8%

$$\underline{85\% \text{ of } 8\% \text{ of T.E.V}}$$

$$0.85 \times 0.08 \times \text{T.E.V}$$

$$\rightarrow 0.068 \text{ of T.E.V}$$

$$\underline{6.8\% \text{ of T.E.V.}}$$

[MCQ]



#Q. A company run a training week for its employees consisting 24 managers and 60 sales people. If 25% of the managers and 25% of sales staff attended the event, then what percent of employees didn't attend the event?

- A** 75%
- B** 12.75%
- C** 25%
- D** 50%

Absentees

$$M \rightarrow \frac{3}{4} \times 24 = 18$$

$$S \rightarrow \frac{3}{4} \times 60 = 45$$

$$\begin{array}{r} 2+3 \\ 63 \\ \hline 84 \\ 28 \\ \hline 112 \end{array} \times \frac{25}{100} = 75$$

[MCQ]



#Q. If the length of a rectangle increase by 20%, then by how much percent we should decrease the breadth in order to maintain same area?

- A** 20%
- B** 16.67%
- C** 83.33%
- D** 81.33%

$$16.67\% \downarrow$$

$$\frac{100}{120} = \frac{5}{6}$$

$$0.833 = 0.83\bar{3}$$



$$A = l \times b$$
$$\frac{120}{100}$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

[MCQ]



#Q. If the side of a square is increased by 20%, then what is the percentage change in its area?

- A** 44%
- B** 80%
- C** 22%
- D** 144%

$$\begin{aligned} A &= S \times S \\ 1.2 \times 1.2 \\ &= 1.44 \end{aligned}$$

44% ↑

PROFIT & LOSS



Profit / Loss

Discount $\rightarrow$ M.P.

PROFIT & LOSS

$$\underline{S.P.} > \underline{C.P.}$$

= Profit

$$\frac{S.P.}{C.P.} > 1$$

$$\underline{S.P.} < \underline{C.P.}$$

= Loss

$$\frac{S.P.}{C.P.} < 1$$

$$\underline{S.P.} = \underline{C.P.}$$

= No Profit No Loss

$$\frac{S.P.}{C.P.} = 1$$

PROFIT & LOSS

Note:

Profit or loss percentage is to be applied always to the Cost Price only.

Discount percentage is to be applied always to the Marked Price only.

✱ A trader may sometimes have multiple profits or losses simultaneously.

This is equivalent to having multiple changes and so all individual changes are to be multiplied to get the overall effect.

[MCQ]



#Q. In what proportion must water be mixed with milk to gain $16\frac{2}{3}\%$ by selling it at cost price?

A

2:3

B

3:2

C

1:6

D

1:3

7 Mixture
6 Milk

$$\frac{\text{Mixture}}{\text{Milk}} = \frac{\text{S.P.}}{\text{C.P.}}$$

350
300

16.66%

Invest Milk

Water : Milk = 1:6

350
300

[MCQ]



#Q. The price of an article is raised by 40% of its cost price and is then sold at a 25% discount for ₹ 3570. What is the cost price of the article?

S.P

$$M.P. = 1.4 \underline{C.P.}$$

$$3570 = 0.75 \times 1.4 C.P.$$

$$\Rightarrow \frac{3570}{1.05} = C.P.$$

$$C.P = 3400$$

A ₹ 3270

B ₹ 3400

C ₹ 3200

D ₹ 3600

[MCQ]



#Q. A shopkeeper advertises for selling cloth at 4% loss. However by using a false meter scale he actually gains 25%. What is actual length of scale?

$$\frac{S.P}{C.P} \rightarrow 1.25 = 0.96 \times \frac{100}{x}$$

$$\therefore x = \frac{96}{1.25} = 76.8 \text{ cm}$$

A 76.8 cm

B 77.8 cm

C 74.8 cm

D 75.8 cm

[MCQ]



#Q. A trader sold a rice bag at a profit of 10%. If he had purchased it for 10% less and sold it for ₹31.20 less, he would have made a profit of 15%. What is the cost price of the rice bag?

$$\frac{S.P}{C.P} = 1.1 \Rightarrow S.P = 1.1 C.P$$

A ₹ 240

B ₹ 320

C ₹ 500

D ₹ 480

$$\frac{S.P - 31.2}{0.9 C.P} = 1.15$$

$$\Rightarrow 1.1 C.P - 31.2 = 1.035 C.P$$

$$\Rightarrow 1.1 C.P - 1.035 C.P = 31.2$$

$$0.065 C.P = 31.2$$

$$C.P = \frac{31.2}{0.065}$$

$$= 480$$

[MCQ]



#Q. A trader marks his goods in advance at 80% more than the cost price, but he allows 15 articles to the dozen and also 10% discount for cash. What rate of overall profit does he obtain from a customer who pays cash?

A 18.2%

B 12.5%

C 32.5%

D 29.6%

$$\underline{M.P} = 1.8 \underline{C.P.}$$

$$\frac{12^4}{15} \times 0.9 \times 1.8 \underline{C.P.} = S.P$$

$$29.6$$

$$29.6\%$$

$$\frac{S.P}{C.P} = 0.8 \times 0.9 \times 1.8 \quad \bigg| \quad \frac{S.P}{C.P} = 1.296$$
$$= 0.72 \times 1.8$$

[MCQ]



#Q. Price of an item is increased by 20% of its cost price and is then sold at 10% discount for ₹ 2160. What is its cost price?

$$0.9 \times 1.2 \text{ C.P} = 2160$$

$$\underline{\text{C.P}} = \frac{2160}{1.08} = 2000$$

A ₹ 1680

B ₹ 1700

C ₹ 1980

D ₹ 2000

[MCQ]

#Q. Three comparable brands of 1 litre cans of a liquid detergent are available in a shop with different offers as shown in the table.

Brand	List price (in ₹ per can)	Offer
<u>A</u>	320	1 / 3rd extra
B	332	1 free for 3
C	<u>300</u>	20% discount

If 4 litres of detergent is to be purchased, then the best choice (based on unit price) would be

$$\begin{aligned}
 \checkmark A &\Rightarrow \textcircled{3 \text{ lit}} + \underline{1 \text{ lit}} \Rightarrow 320 \times 3 = 960 \rightarrow \frac{960}{4} = \textcircled{240} \\
 B &\Rightarrow \textcircled{3 \text{ lit}} + \underline{1 \text{ lit}} \Rightarrow 332 \times 3 = 996 \rightarrow \frac{996}{4} = \underline{249} \\
 \checkmark C &\Rightarrow 300 \times 0.8 = \textcircled{240}
 \end{aligned}$$

- ☒ **A** A or C
- ☐ **B** A or B
- ☐ **C** B or C
- ☐ **D** B

[MCQ]



#Q. The wholesale price per unit of an item is C_0 up to first 19 units. The unit price falls by 10% if 20 to 29 units are purchased, and by another 10% if 30 or more units are purchased. If 120 units are bought, the unit price paid is

- A** $0.89 C_0$
- B** $0.97 C_0$
- C** $0.91 C_0$
- D** $0.81 C_0$

$$0.9 \times 0.9 \times C_0$$

$$0.81 C_0$$



2 mins Summary



Topic

Percentages, Profit-Loss & ~~Mixtures-Alligations~~



THANK - YOU

